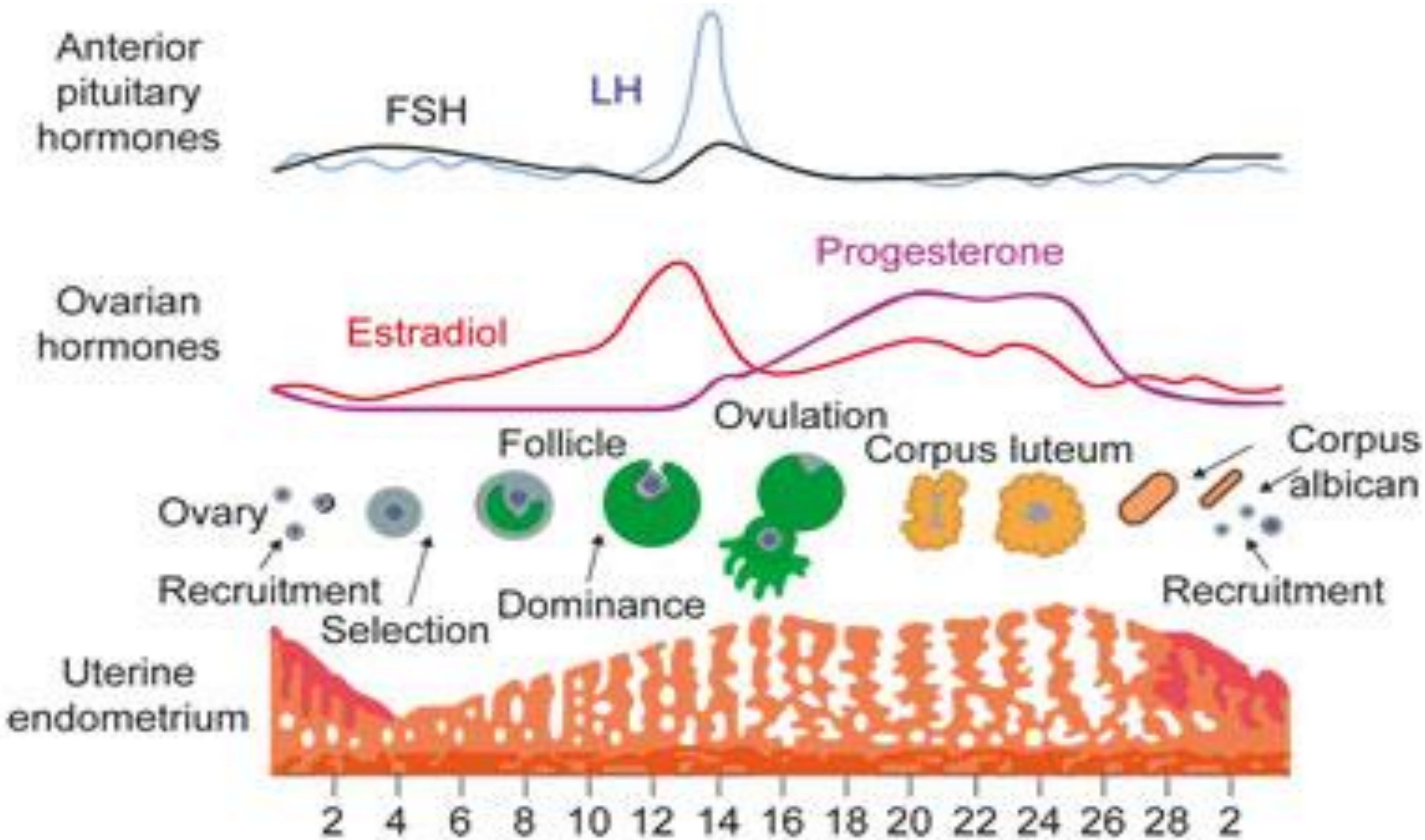
Abnormal uterine bleeding (AUB)

Prepared by
dr. REZAN ABD-UL-RAHMAN AL-UTHMAN
M.B.Ch.B ,F.I.B.O.G
2025

Menstruation is a woman's monthly bleeding from her reproductive tract induced by hormonal changes of the menstrual cycle. The length of a menstrual cycle is the time from the start of a period to the start of the next.



Terms to be kept in new terminology	
Abnormal uterine bleeding (AUB)	Any menstrual bleeding from the uterus that is either abnormal in volume (excessive duration or heavy), regularity, timing (delayed or frequent) or is non-menstrual (IMB, PCB or PMB)
Heavy menstrual bleeding (HMB)	For clinical purposes, HMB is defined as excessive menstrual blood loss leading to interference with the physical, emotional, social and material quality of life of a woman, and which can occur alone or in combination with other symptoms. Adverse outcome is greater in women with total MBL that exceeds 80 ml or menses duration >7 days
Intermenstrual bleeding (IMB*)*	Uterine bleeding that occurs between clearly defined cyclic and predictable menses. Such bleeding may occur at random times or may manifest in a predictable fashion at the same day in each cycle

Postmenopausal bleeding (PMB)	Genital tract bleeding that recurs in a menopausal woman at least one year after cessation of cycles
Postcoital bleeding (PCB)	Non-menstrual genital tract bleeding immediately (or shortly after) intercourse
Chronic AUB	AUB has been present for the majority of the past 6 months. In most cases, chronic AUB is unlikely to require urgent immediate clinical intervention
Acute AUB	Excessive AUB bleeding that requires immediate intervention to prevent further blood loss. Acute AUB may present in the context of existing chronic AUB or might occur without such a history

Terms that should be discarded in new terminology

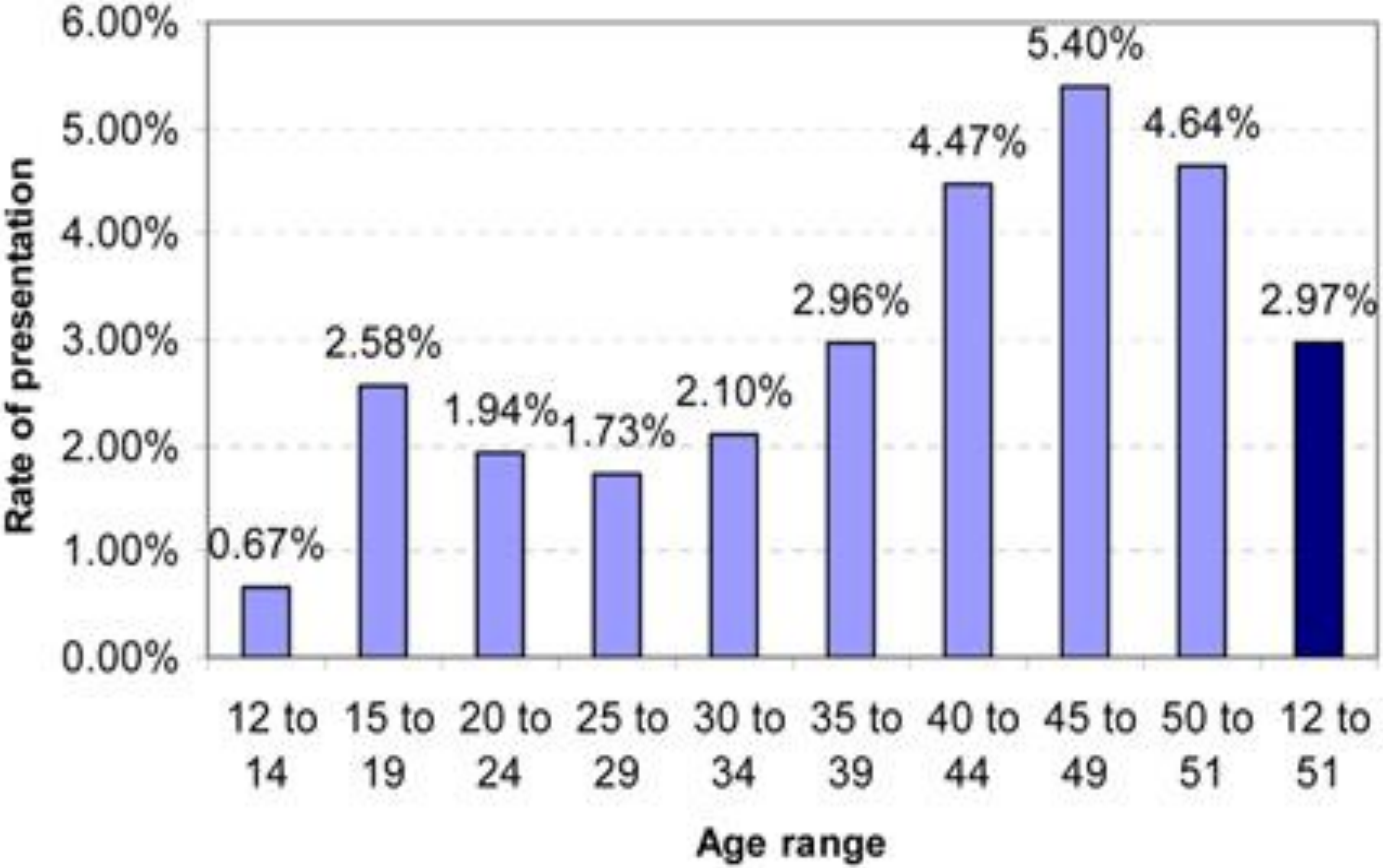
Menorrhagia	Heavy menstrual bleeding that occurs at expected intervals of the menstrual cycle (cycle length varying from 21 to 35 days)
Oligomenorrhoea	Bleeding occurs at intervals of >35 days and <6 months, usually caused by a prolonged follicular phase
Polymenorrhoea	Regular bleeding at intervals of less than 3 weeks, which may be caused by a luteal phase defect
Amenorrhoea	No uterine bleeding for at least 6 months

Definition of HMB

- Heavy menstrual bleeding (HMB) is clinically defined as menstrual blood loss (MBL) that is subjectively considered to be excessive by the woman and interferes with her physical, emotional, social and material quality of life.
- Quantifying monthly menstrual blood loss (MBL) does not improve clinical care and is not undertaken in modern clinical practice.

- Objective assessment: a PBAC(pictorial blood loss assessment chart) score greater than or equal to 100 equates to a sensitivity of 86%,-91% and a specificity of 82–89% in predicting MBL greater than 80 ml by alkaline hematin test (gold standard reference technique).
- Subjective assessment: subjective assessment of MBL combines information of sanitary protection usage, flooding, clots, duration of menstruation and the woman's personal opinion of her menstrual loss. Although this tends to be inaccurate, it is easy to undertake in clinical practice and is the preferred method of assessing HMB

Annual rate of women with HMB presenting to service



Prevalence of HMB

- HMB has a major adverse effect on the quality of life of many women. Overall, 3% of premenopausal women experience HMB. However, this absolute risk is almost doubled in woman aged 40–51. HMB accounts for around 15% of all secondary care gynaecological referrals in the UK.

Causes of HMB

- Between 40–60% of women with HMB have no uterine, endocrine, hematological or infective pathology on investigation. These women were formerly termed to have dysfunctional uterine bleeding (DUB) of ovulatory (regular cycle) or anovulatory (irregular cycle) type.
- Most HMB is due to a combination of coagulopathy, ovulatory or endometrial dysfunction that does not require secondary referral and treatment can be commenced in primary care. The term DUB should be discarded and PALM-COEIN classification adopted.

Abnormal Uterine Bleeding (AUB)

PALM (structural causes)

- Polyp
- Adenomyosis
- Leiomyoma
  - Submucosal myoma
  - Other myoma
- Malignancy and hyperplasia

COEIN (nonstructural causes)

- Coagulopathy
- Ovulatory dysfunction
- Endometrial
- Iatrogenic
- Not yet classified

PALM-COEIN classification

Category	**Number of patients n=300 (%)**
Polyp (P)	8(2.66)
Adenomyosis (A)	28 (9.33)
Leiomyoma (L)	68 (22.66)
Malignancy (M)	8 (2.66)
Coagulopathy (C)	3 (1)
Ovulatory dysfunction (O)	85 (28.33)
Endometrial (E)	62(20.66)
Iatrogenic (I)	13 (4.33)
Not yet classified (N)	25 (8.33)

- Gynecological malignancy rarely presents as HMB, but can present as prolonged intermenstrual bleeding (IMB), postcoital bleeding (PCB), postmenopausal bleeding (PMB) and as a pelvic mass

Assessment of AUB - generic principles

Establish if chronic AUB (>6 months) or acute AUB (urgent intervention required)

Initial step should be to exclude pregnancy

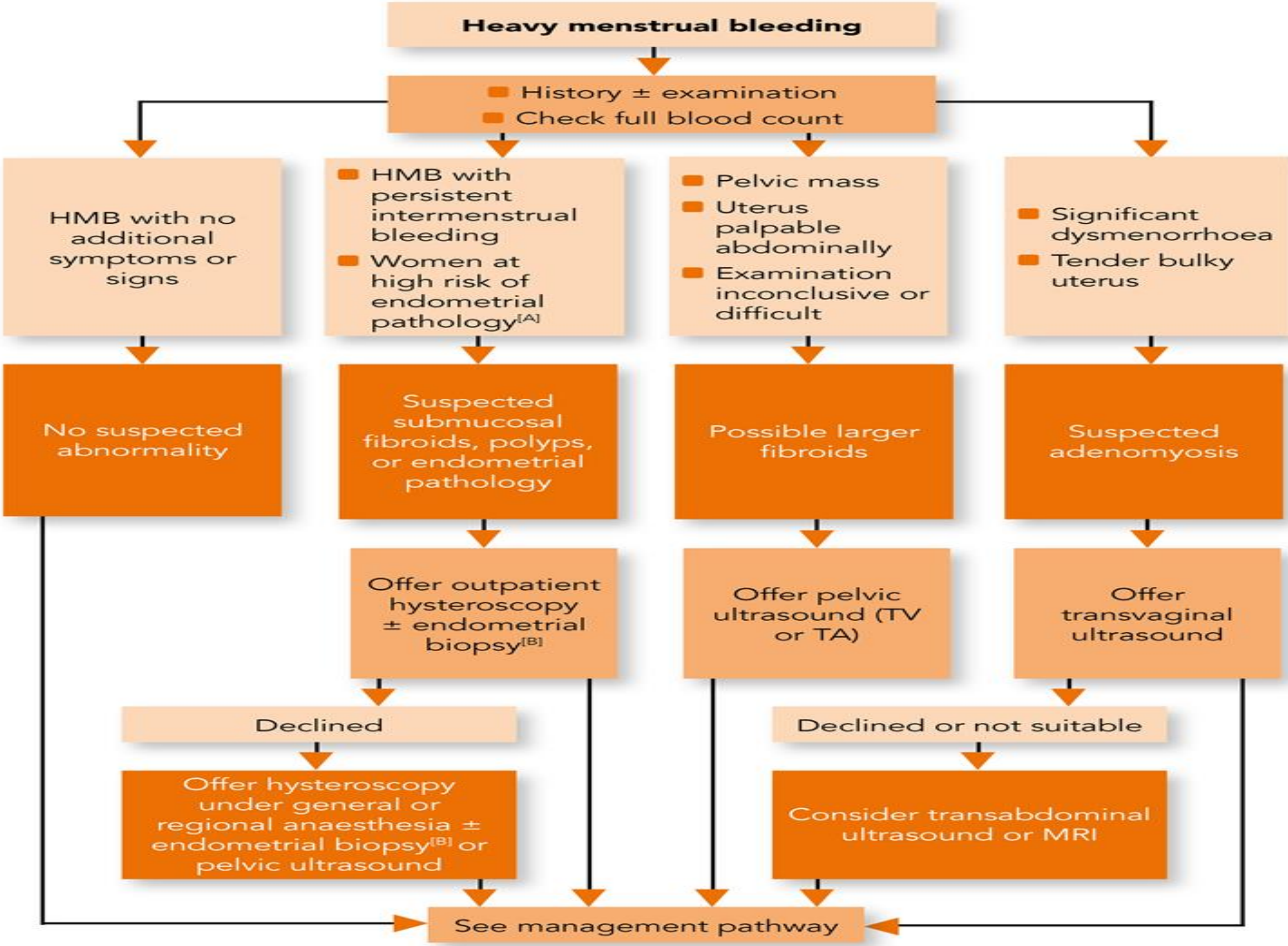
Undertake relevant gynecological history and examination

- Recommendation for full blood count (FBC), cervical smear, pelvic infection swabs and pelvic ultrasound (and coagulation screen if clinical question screen positive

- Screen for coagulopathy (such as von Willebrand disease) by using a specific structured history. If clinical history screen is positive then undertake coagulation profile and vWD test

- Referral to secondary care if malignancy is suspected, endometrial biopsy is required, pathology suspected/identified or medical treatment deemed unsuccessful.

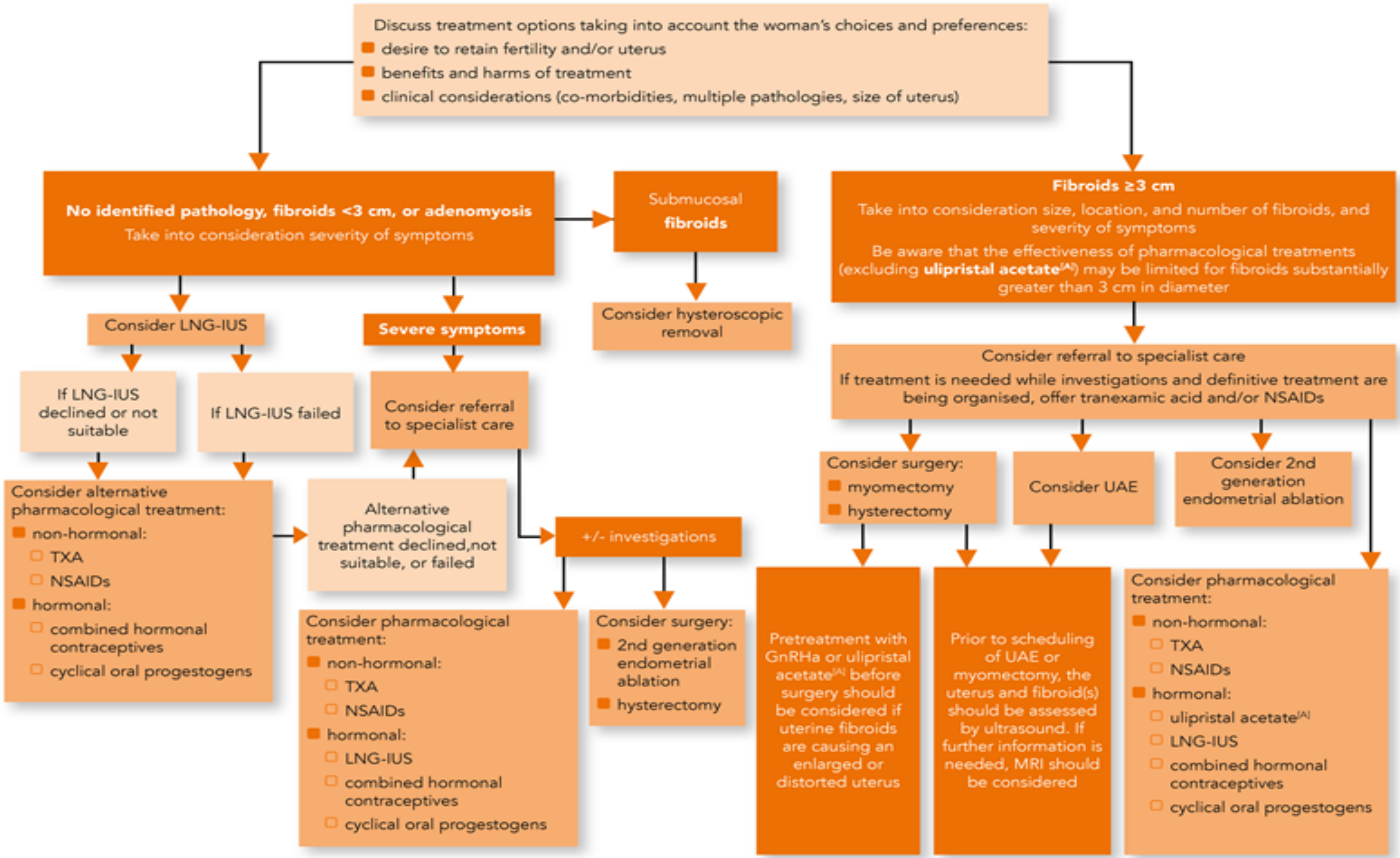
- Most HMB is due to a combination of coagulopathy, ovulatory or endometrial dysfunction that does not require secondary referral and treatment can be commenced in primary care
- Uterine/endometrial assessment: investigation using hysteroscopy with endometrial biopsy or ultrasound and endometrial biopsy improves the detection rate of endometrial pathology (malignant and benign) compared with hysteroscopy or ultrasound alone



HMB=heavy menstrual bleeding; MRI=magnetic resonance imaging; TA=transabdominal; TV=transvaginal

[A] For example, women with persistent intermenstrual or persistent irregular bleeding and women with infrequent bleeding who are obese or have polycystic ovary syndrome, women taking tamoxifen, women for whom treatment for HMB has been unsuccessful

[B] If high risk for endometrial pathology (see footnote A)

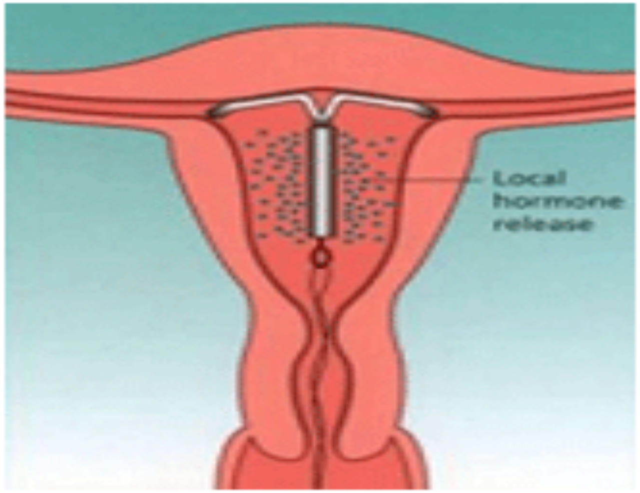


GnRHa=gonadotrophin-releasing hormone analogue; LNG-IUS=levonorgestrel-releasing intrauterine system; NSAIDs=non-steroidal anti-inflammatory drugs; TXA=tranexamic acid; UAE=uterine artery embolisation.

[A] Ulipristal acetate is indicated for one course (up to 3 months) of preoperative treatment for moderate to severe symptoms of uterine fibroids and for intermittent treatment (up to four courses) of moderate to severe symptoms of uterine fibroids in women ineligible for surgery. See also the summary of product characteristics.

Levonorgestrel-releasing intrauterine system (LNG-IUS)

- The levonorgestrel intrauterine-releasing system (LNG-IUS) is an intrauterine, long-term progestogen-only method of contraception currently licensed for 5 years of use.
- It has T-shaped plastic frame with a rate-limiting membrane on the vertical stem, releasing a daily intrauterine dose of 20 microgram/24 hours of levonorgestrel with little systemic absorption. The LNG-IUS inhibits endometrial proliferation, thickens cervical mucus and suppresses ovulation.



Endometrial resection and endometrial ablation

- There are several minimally invasive procedures to destroy the endometrium in premenopausal women with menorrhagia due to benign causes for whom childbearing is complete.
- Traditionally, first-generation techniques involved resection (transcervical resection of the endometrium [TCRE]) or ablation (rollerball) of the endometrium under direct hysteroscopic vision using electrocautery (either monopolar or bipolar).

- Short-term effectiveness (up to 2 years) is high for first- and second-generation endometrial destruction: around 25–35% of women treated experience amenorrhoea, with over 90% experiencing satisfactory reduction in menstrual blood loss without the need for further treatment. And 95% of women return to normal activities by 2 weeks.

- However, upon longer follow up (around 5 years), around 20–30% of women opting for ablation will either be dissatisfied or require secondary treatment (usually hysterectomy).
- All techniques are performed as day-case procedures with some being able to be performed under local anaesthetic (with or without sedation) in an outpatient setting.
- There is evidence to suggest that benefits result from thinning the endometrium (with medical therapy such as GnRHa) prior to ablation, or performing the ablation just after menses is complete


NovaSure


2
1

Women suitable for endometrial ablation

- Endometrial ablation (uterus-conserving) and hysterectomy are treatment options offered to women with unsuccessful medical treatment (including Mirena LNG-IUS) for their menorrhagia. Informed consent is required that provides the individualized risks and benefits of the treatment against other potential alternatives.
- However, endometrial ablation is best suited to women whose:
- uterus is not greater than 10 cm in size (although microwave ablation may be used in uteri up to 14 cm cavity length)

- uterus does not contain large fibroids that may be distorting the uterine cavity
- intrauterine cavity does not contain any polyps or fibroids greater than 3 cm, otherwise these should be removed prior to ablation
- uterus has not undergone any previous endometrial ablation procedure, active infective process and whose myometrium is at least 10 mm if using the microwave ablation technique
- family is complete or they have no desire for future fertility

Recommendations prior to endometrial ablation

- Endometrial biopsy is obtained and histologically analysed to exclude the possibility of endometrial hyperplasia or endometrial cancer.
- Hysteroscopy is performed immediately prior to the insertion of the ablation device to ensure that:
- any sounding or dilation of the cervix has not caused a perforation or false passage, resulting in subsequent introduction of the ablation device into the wrong space
- there is no significant intrauterine pathology that would preclude ablation, e.g. uterine cavity distorted by intrauterine fibroids
- the operator can accurately determine uterine cavity length (and cervical canal length if using NovaSure®

Hysterectomy

Vaginal hysterectomy

Individual patient assessment is essential when deciding the route of hysterectomy, and the following factors need to be taken into account:

- presence of other gynecological conditions or disease
- uterine size
- presence and size of uterine fibroids
- mobility and descent of uterus
- size and shape of vagina
- history of previous surgery
- There is insufficient evidence on the safety and role of total laparoscopic hysterectomy (NICE guidance). However, laparoscopic-assisted vaginal hysterectomy may be offered to the woman provided she has been informed of the alternatives and their risks/benefits.

Abdominal hysterectomy

Preoperative GnRHa treatment for 3–4 months may facilitate surgery by reducing preoperative anemia, intraoperative blood loss and the need for transfusion. Studies have shown a reduction of fibroid size of by 60%, which may enable a lower transverse incision (lower morbidity) rather than vertical midline laparotomy incision (higher morbidity) when conducting the abdominal hysterectomy.

Indications

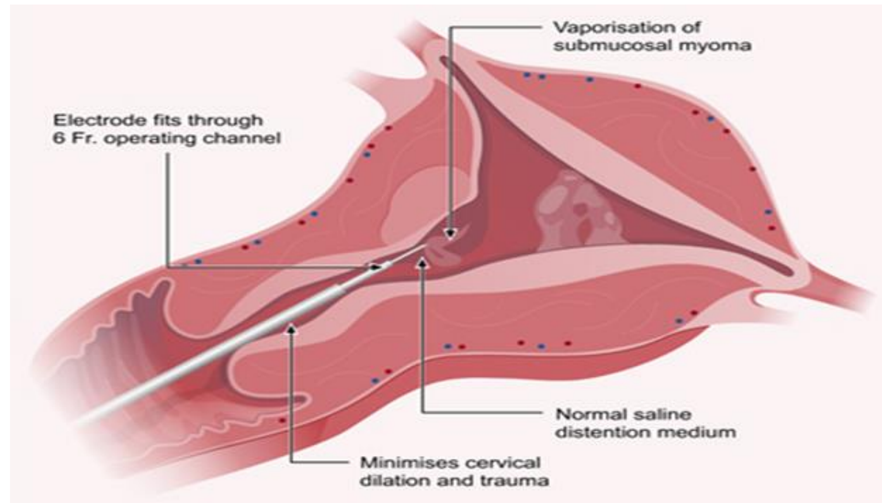
Hysterectomy should be only considered where:

- other treatment options have failed or are inappropriate
- women have completed their families
- there is a wish for amenorrhea
- women (who have been fully counselled) request it or other forms of further treatment are contraindicated

Polyps arising from the uterine endometrium

- Endometrial polyps are localized hyperplastic overgrowths of endometrial glands and stroma which form projections over the endometrial surface.
- Aetiology remains obscure, however it is proposed that, polyps lose their apoptotic cell regulation and overexpress estrogen and progesterone receptors causing them to grow.
- Endometrial polyps are rare before the age of 20 with the incidence steadily rising after that, peaking in the 5th decade of life and then declining at menopause. Incidence estimates are variable based on the population studied, however, 10% of asymptomatic women and 24-41% of women with AUB (Abnormal uterine bleeding) have endometrial polyps.

Clinical presentation
Asymptomatic Heavy menstrual bleeding
Postmenopausal bleeding
Prolapse through cervical ostium
Abnormal vaginal discharge,
Breakthrough bleeding, Infertility



Large or multiple endometrial polyps can contribute to miscarriage and infertility. Evidence also suggests that a polypectomy may improve spontaneous conception rates in women with an endometrial polyp as the only factor contributing to subfertility by normalising endometrial implantation factors•

Polyp Continue….

Malignancy

- Polyp size of >1 cm, abnormal uterine bleeding and postmenopausal status are all independent risk factors for malignant polyps. Hysteroscopic markers for malignant endometrial polyps include surface irregularities such as necrosis, vascular irregularities and whitish thickened areas, which are indications for obtaining a histological diagnosis.

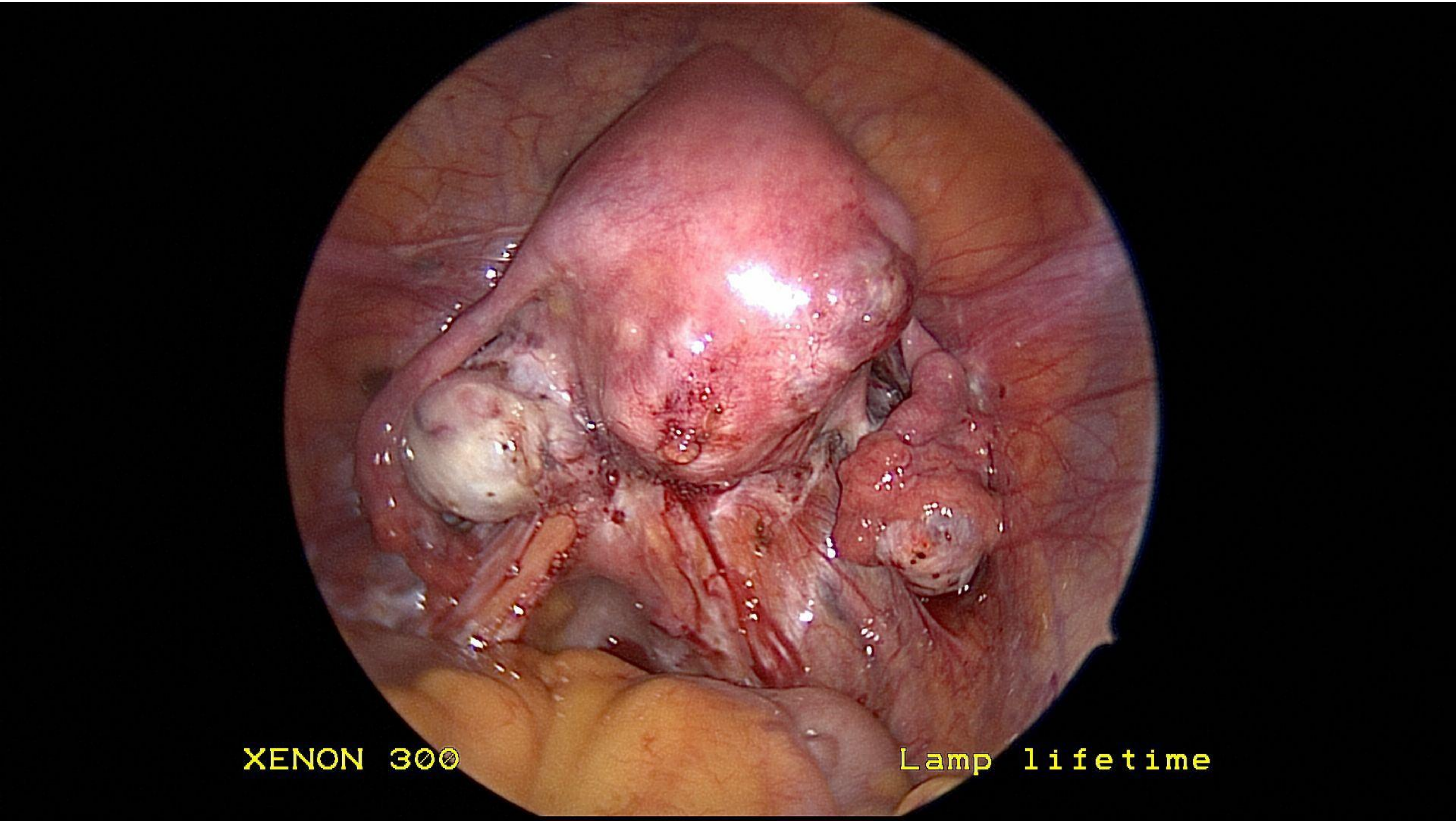
Diagnostic modalities

- Pelvic ultrasound
- Saline infusion sonogram (SIS) – Safe, well tolerated, rapid, minimally invasive, highly sensitive and superior to ultrasound in diagnosing polyps
- Hysteroscopy – Gold standard. Can be used to see and treat at the same sitting.

Adenomyosis

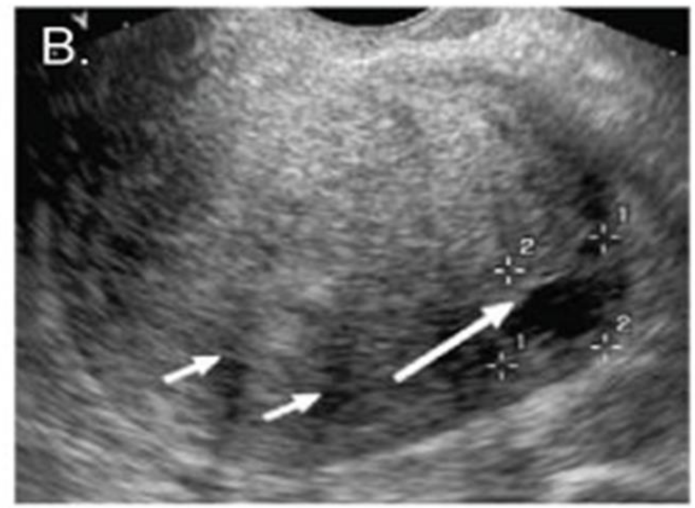
Adenomyosis is a benign, common gynecological condition causing heavy, painful periods in premenopausal women who tend to be multiparous and between 40 and 50 years of age. Overall it is considered to contribute to approximately 10% of all cases of HMB and 30% of all cases of HMB with dysmenorrhea.

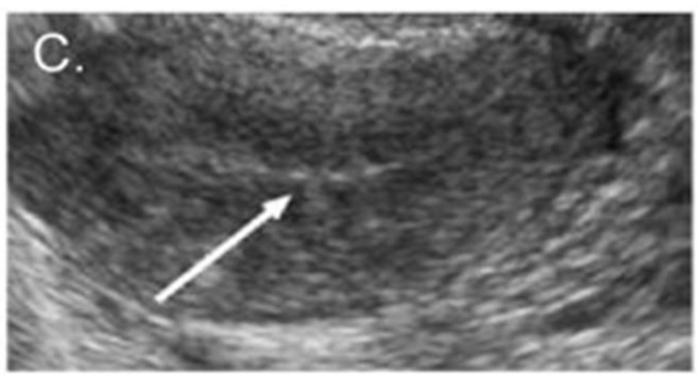
- It is defined histologically (usually on a hysterectomy specimen) as the presence of non-neoplastic endometrial glands and stroma in the myometrium. It is often associated with hypertrophy and hyperplasia of the myometrium surrounding the ectopic endometrial tissue. However, the clinical diagnosis of adenomyosis prior to hysterectomy is often inaccurate. This is due to:
- variability and non-specific nature of symptoms, which tends to mimic any combination of uterine fibroids, endometriosis and HMB due to ovulatory or endometrial dysregulation
- variability of the sensitivity and specificity of pelvic ultrasound to diagnose adenomyosis.


XENON 300
Lamp lifetime

Adenomyosis Screening investigations

- Transvaginal ultrasound is useful as a screening tool in most patients by and an experienced sonographer .
- However, pelvic ultrasound becomes less predictive for adenomyosis in women with co-existing fibroids. MRI has a higher diagnostic capability, irrespective of the presence or absence of uterine fibroids.





Pelvic ultrasound images depicting adenomyosis

Figure A. Myometrial heterogeneity (arrows)

Figure B. Myometrial linear striations (short arrows) and a myometrial cyst (long arrow)

Figure C. Heterogeneous myometrium with irregular endo/myometrial interface (arrow)

Clinical treatment

- Management of adenomyosis uteri is hindered by the lack of a reliable, noninvasive diagnostic test. No serum markers are currently available.

Medical:

- Nonhormonal therapy, including mefenamic and tranexamic acid, may be effective for the symptomatic relief of menorrhagia
- Low-dose, continuous combined oral contraceptives with withdrawal bleeds every 4-6 months may be effective in relieving menorrhagia and dysmenorrhea
- high-dose continuous daily oral progestogens
- GnRHa agonist
- Mirena LNG-IUS
- Uterus conserving:
- Balloon endometrial ablation

- Uterus-conserving minimally invasive treatments: uterine artery embolisation.
- Adenomyoma excision at the time of abdominal myomectomy
- Laparoscopic myometrial electrocoagulation induces localised coagulation and necrosis of adenomyosis uteri. This technique may be used in women > 40 years of age who have completed their families, but who wish to avoid hysterectomy. Risks include adhesion formation, bleeding resulting in a hysterectomy and difficulty in precise application resulting in the weakening of the remainder of the myometrial tissue.
- Hysterectomy
- New treatments for adenomyosis being trialled include: aromatase inhibitors, selective progesterone receptor modulators (e.g. asoprisnil) and bipolar radiofrequency ablation via hysteroscopy or laparoscopy

Thank you